

```

import { useState, useEffect, useRef, useCallback } from "react";

const DRILLS = [
  {
    id: 1,
    name: "Triangle Passing",
    pillar: "Passing",
    pillarColor: "#3B82F6",
    duration: "12 min",
    players: "Groups of 3",
    focus: "Short passing & movement",
    cue: "Pass and move – never stand still!",
    steps: ["A passes to B", "A follows pass to B's cone", "B passes to C", "B fo
  },
  {
    id: 2,
    name: "Cone Slalom Dribbling",
    pillar: "Dribbling",
    pillarColor: "#22C55E",
    duration: "12 min",
    players: "Full squad",
    focus: "Ball control at speed",
    cue: "Head up – feel the ball!",
    steps: ["Player starts at cone 1", "Weave through cones using inside/outside
  },
  {
    id: 3,
    name: "1v1 Shooting",
    pillar: "Finishing",
    pillarColor: "#EF4444",
    duration: "15 min",
    players: "Groups of 4",
    focus: "Finishing under pressure",
    cue: "Stay low, use feints, shoot early!",
    steps: ["Coach plays ball to attacker", "Attacker drives at defender", "Attac
  },
  {
    id: 4,
    name: "4v2 Rondo",
    pillar: "Passing",
    pillarColor: "#3B82F6",
    duration: "12 min",
    players: "Groups of 6",
    focus: "Keeping ball under pressure",

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    cue: "Move after every pass – create angles!",
    steps: ["Outside players keep possession", "Defenders try to intercept", "Bal
},
{
    id: 5,
    name: "Gate Dribbling",
    pillar: "Dribbling",
    pillarColor: "#22C55E",
    duration: "10 min",
    players: "Full squad",
    focus: "Ball control & direction changes",
    cue: "Head up – find the next gate!",
    steps: ["Player starts dribbling freely", "Spot an open gate", "Drive through
},
{
    id: 6,
    name: "2v1 to Goal",
    pillar: "Finishing",
    pillarColor: "#EF4444",
    duration: "12 min",
    players: "Groups of 4",
    focus: "Combination play & overlaps",
    cue: "Commit the defender, then release!",
    steps: ["A1 drives at the defender", "A2 makes overlapping run wide", "Defend
},
{
    id: 7,
    name: "Crossing & Finishing",
    pillar: "Finishing",
    pillarColor: "#EF4444",
    duration: "15 min",
    players: "Groups of 4",
    focus: "Delivery and attacking the box",
    cue: "Cross early – striker, attack the near post!",
    steps: ["Winger drives from cone down the flank", "Striker makes near post ru
},
{
    id: 8,
    name: "Shadow Defending",
    pillar: "Defending",
    pillarColor: "#F97316",
    duration: "10 min",
    players: "Pairs",
    focus: "Positioning & patience",
    cue: "Stay goal-side – bend your knees – be patient!",
    steps: ["Attacker moves freely in the area", "Defender tracks – stays between
},

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{
  id: 9,
  name: "3v2 Counter Attack",
  pillar: "Finishing",
  pillarColor: "#EF4444",
  duration: "15 min",
  players: "Groups of 5",
  focus: "Quick attacking transitions",
  cue: "Find the free player – attack the space!",
  steps: ["3 attackers launch from halfway", "A1 carries ball forward", "Defend
},
{
  id: 10,
  name: "Switch of Play",
  pillar: "Passing",
  pillarColor: "#3B82F6",
  duration: "12 min",
  players: "Groups of 6+",
  focus: "Vision, weight of pass & width",
  cue: "Play where they are NOT!",
  steps: ["Ball starts on congested right side", "CM receives under pressure",
},
];

```

// — Animation data for each drill

```

function getDrillAnimation(id) {
  switch (id) {
    case 1: // Triangle Passing
      return {
        fieldType: "mini",
        keyframes: [
          {
            t: 0,
            players: [
              { id: "A", x: 50, y: 75, color: "#3B82F6", label: "A" },
              { id: "B", x: 20, y: 25, color: "#3B82F6", label: "B" },
              { id: "C", x: 80, y: 25, color: "#3B82F6", label: "C" },
            ],
            ball: { x: 50, y: 75 },
            trails: [],
          },
          {
            t: 0.25,
            players: [
              { id: "A", x: 50, y: 75, color: "#3B82F6", label: "A" },
              { id: "B", x: 20, y: 25, color: "#3B82F6", label: "B" },
              { id: "C", x: 80, y: 25, color: "#3B82F6", label: "C" },
            ],
          },
        ],
      };
    }
  }
}

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    ],
    ball: { x: 20, y: 25 },
    trails: [{ x1: 50, y1: 75, x2: 20, y2: 25 }],
  },
  {
    t: 0.42,
    players: [
      { id: "A", x: 20, y: 25, color: "#22C55E", label: "A" },
      { id: "B", x: 20, y: 25, color: "#3B82F6", label: "B" },
      { id: "C", x: 80, y: 25, color: "#3B82F6", label: "C" },
    ],
    ball: { x: 20, y: 25 },
    trails: [],
  },
  {
    t: 0.58,
    players: [
      { id: "A", x: 20, y: 25, color: "#22C55E", label: "A" },
      { id: "B", x: 20, y: 25, color: "#3B82F6", label: "B" },
      { id: "C", x: 80, y: 25, color: "#3B82F6", label: "C" },
    ],
    ball: { x: 80, y: 25 },
    trails: [{ x1: 20, y1: 25, x2: 80, y2: 25 }],
  },
  {
    t: 0.75,
    players: [
      { id: "A", x: 20, y: 25, color: "#22C55E", label: "A" },
      { id: "B", x: 80, y: 25, color: "#22C55E", label: "B" },
      { id: "C", x: 80, y: 25, color: "#3B82F6", label: "C" },
    ],
    ball: { x: 80, y: 25 },
    trails: [],
  },
  {
    t: 0.88,
    players: [
      { id: "A", x: 20, y: 25, color: "#22C55E", label: "A" },
      { id: "B", x: 80, y: 25, color: "#22C55E", label: "B" },
      { id: "C", x: 80, y: 25, color: "#3B82F6", label: "C" },
    ],
    ball: { x: 50, y: 75 },
    trails: [{ x1: 80, y1: 25, x2: 50, y2: 75 }],
  },
  {
    t: 1,
    players: [

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        { id: "A", x: 20, y: 25, color: "#22C55E", label: "A" },
        { id: "B", x: 80, y: 25, color: "#22C55E", label: "B" },
        { id: "C", x: 50, y: 75, color: "#22C55E", label: "C" },
    ],
    ball: { x: 50, y: 75 },
    trails: [],
},
],
staticElements: [
    { type: "cone", x: 50, y: 75 },
    { type: "cone", x: 20, y: 25 },
    { type: "cone", x: 80, y: 25 },
    { type: "dashed-line", x1: 50, y1: 75, x2: 20, y2: 25 },
    { type: "dashed-line", x1: 20, y1: 25, x2: 80, y2: 25 },
    { type: "dashed-line", x1: 80, y1: 25, x2: 50, y2: 75 },
],
};

```

case 2: // Cone Slalom

```

return {
    fieldType: "mini",
    keyframes: (() => {
        const coneXs = [15, 30, 45, 60, 75, 88];
        const coneY = 50;
        const frames = [];
        const pts = [
            { x: 5, y: 50 },
            { x: 15, y: 30 }, { x: 30, y: 70 }, { x: 45, y: 30 },
            { x: 60, y: 70 }, { x: 75, y: 30 }, { x: 88, y: 70 },
            { x: 95, y: 50 },
        ];
        pts.forEach((p, i) => {
            frames.push({
                t: i / (pts.length - 1),
                players: [{ id: "P", x: p.x, y: p.y, color: "#3B82F6", label: "P" } ],
                ball: { x: p.x + 2, y: p.y + 2 },
                trails: i > 0 ? [{ x1: pts[i-1].x, y1: pts[i-1].y, x2: p.x, y2: p.y } ] : [],
            });
        });
        return frames;
    })(),
    staticElements: [
        { type: "cone", x: 15, y: 50 },
        { type: "cone", x: 30, y: 50 },
        { type: "cone", x: 45, y: 50 },
        { type: "cone", x: 60, y: 50 },
        { type: "cone", x: 75, y: 50 },
    ],
};

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        { type: "cone", x: 88, y: 50 },
    ],
};

case 3: // 1v1 Shooting
return {
    fieldType: "half",
    keyframes: [
        { t: 0, players: [
            { id: "A", x: 50, y: 88, color: "#3B82F6", label: "A" },
            { id: "D", x: 50, y: 55, color: "#EF4444", label: "D" },
            { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
        ], ball: { x: 50, y: 88 }, trails: [] },
        { t: 0.3, players: [
            { id: "A", x: 50, y: 68, color: "#3B82F6", label: "A" },
            { id: "D", x: 50, y: 55, color: "#EF4444", label: "D" },
            { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
        ], ball: { x: 50, y: 68 }, trails: [] },
        { t: 0.55, players: [
            { id: "A", x: 38, y: 58, color: "#3B82F6", label: "A" },
            { id: "D", x: 52, y: 52, color: "#EF4444", label: "D" },
            { id: "GK", x: 52, y: 15, color: "#EAB308", label: "GK" },
        ], ball: { x: 38, y: 58 }, trails: [] },
        { t: 0.75, players: [
            { id: "A", x: 38, y: 42, color: "#3B82F6", label: "A" },
            { id: "D", x: 52, y: 52, color: "#EF4444", label: "D" },
            { id: "GK", x: 44, y: 15, color: "#EAB308", label: "GK" },
        ], ball: { x: 38, y: 42 }, trails: [] },
        { t: 1, players: [
            { id: "A", x: 38, y: 42, color: "#3B82F6", label: "A" },
            { id: "D", x: 52, y: 52, color: "#EF4444", label: "D" },
            { id: "GK", x: 44, y: 15, color: "#EAB308", label: "GK" },
        ], ball: { x: 38, y: 8 }, trails: [{ x1: 38, y1: 42, x2: 38, y2: 8, col
    ],
    staticElements: [
        { type: "goal", x: 50, y: 5 },
        { type: "cone", x: 50, y: 92 },
    ],
};

case 4: // 4v2 Rondo
return {
    fieldType: "mini",
    keyframes: [
        { t: 0, players: [
            { id: "O1", x: 15, y: 20, color: "#3B82F6", label: "O" },
            { id: "O2", x: 85, y: 20, color: "#3B82F6", label: "O" },

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    { id: "O3", x: 85, y: 80, color: "#3B82F6", label: "O" },
    { id: "O4", x: 15, y: 80, color: "#3B82F6", label: "O" },
    { id: "D1", x: 38, y: 45, color: "#EF4444", label: "D" },
    { id: "D2", x: 58, y: 55, color: "#EF4444", label: "D" },
  ], ball: { x: 15, y: 20 }, trails: [] },
  { t: 0.2, players: [
    { id: "O1", x: 15, y: 20, color: "#3B82F6", label: "O" },
    { id: "O2", x: 85, y: 20, color: "#3B82F6", label: "O" },
    { id: "O3", x: 85, y: 80, color: "#3B82F6", label: "O" },
    { id: "O4", x: 15, y: 80, color: "#3B82F6", label: "O" },
    { id: "D1", x: 48, y: 25, color: "#EF4444", label: "D" },
    { id: "D2", x: 58, y: 55, color: "#EF4444", label: "D" },
  ], ball: { x: 85, y: 20 }, trails: [{ x1: 15, y1: 20, x2: 85, y2: 20 }]
  { t: 0.4, players: [
    { id: "O1", x: 15, y: 20, color: "#3B82F6", label: "O" },
    { id: "O2", x: 85, y: 20, color: "#3B82F6", label: "O" },
    { id: "O3", x: 85, y: 80, color: "#3B82F6", label: "O" },
    { id: "O4", x: 15, y: 80, color: "#3B82F6", label: "O" },
    { id: "D1", x: 72, y: 42, color: "#EF4444", label: "D" },
    { id: "D2", x: 68, y: 62, color: "#EF4444", label: "D" },
  ], ball: { x: 85, y: 80 }, trails: [{ x1: 85, y1: 20, x2: 85, y2: 80 }]
  { t: 0.6, players: [
    { id: "O1", x: 15, y: 20, color: "#3B82F6", label: "O" },
    { id: "O2", x: 85, y: 20, color: "#3B82F6", label: "O" },
    { id: "O3", x: 85, y: 80, color: "#3B82F6", label: "O" },
    { id: "O4", x: 15, y: 80, color: "#3B82F6", label: "O" },
    { id: "D1", x: 52, y: 72, color: "#EF4444", label: "D" },
    { id: "D2", x: 38, y: 65, color: "#EF4444", label: "D" },
  ], ball: { x: 15, y: 80 }, trails: [{ x1: 85, y1: 80, x2: 15, y2: 80 }]
  { t: 0.8, players: [
    { id: "O1", x: 15, y: 20, color: "#3B82F6", label: "O" },
    { id: "O2", x: 85, y: 20, color: "#3B82F6", label: "O" },
    { id: "O3", x: 85, y: 80, color: "#3B82F6", label: "O" },
    { id: "O4", x: 15, y: 80, color: "#3B82F6", label: "O" },
    { id: "D1", x: 28, y: 48, color: "#EF4444", label: "D" },
    { id: "D2", x: 38, y: 38, color: "#EF4444", label: "D" },
  ], ball: { x: 15, y: 20 }, trails: [{ x1: 15, y1: 80, x2: 15, y2: 20 }]
  { t: 1, players: [
    { id: "O1", x: 15, y: 20, color: "#3B82F6", label: "O" },
    { id: "O2", x: 85, y: 20, color: "#3B82F6", label: "O" },
    { id: "O3", x: 85, y: 80, color: "#3B82F6", label: "O" },
    { id: "O4", x: 15, y: 80, color: "#3B82F6", label: "O" },
    { id: "D1", x: 38, y: 45, color: "#EF4444", label: "D" },
    { id: "D2", x: 58, y: 55, color: "#EF4444", label: "D" },
  ], ball: { x: 85, y: 20 }, trails: [{ x1: 15, y1: 20, x2: 85, y2: 20 }]
  ],
  staticElements: [

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```

    { type: "box", x: 15, y: 20, w: 70, h: 60 },
  ],
};

```

case 5: // Gate Dribbling

```

return {
  fieldType: "mini",
  keyframes: (() => {
    const gates = [{ x: 22, y: 30 }, { x: 55, y: 20 }, { x: 78, y: 40 }, {
    const path = [{ x: 8, y: 85 }, ...gates, { x: 8, y: 85 }];
    return path.map((p, i) => ({
      t: i / (path.length - 1),
      players: [{ id: "P", x: p.x, y: p.y, color: "#22C55E", label: "P" }],
      ball: { x: p.x + 2, y: p.y },
      trails: i > 0 ? [{ x1: path[i-1].x, y1: path[i-1].y, x2: p.x, y2: p.y
    }));
  })(),
  staticElements: [
    { type: "gate", x: 22, y: 30 }, { type: "gate", x: 55, y: 20 },
    { type: "gate", x: 78, y: 40 }, { type: "gate", x: 35, y: 65 },
    { type: "gate", x: 65, y: 72 }, { type: "gate", x: 82, y: 58 },
  ],
};

```

case 6: // 2v1 to Goal

```

return {
  fieldType: "half",
  keyframes: [
    { t: 0, players: [
      { id: "A1", x: 38, y: 85, color: "#3B82F6", label: "A1" },
      { id: "A2", x: 62, y: 85, color: "#3B82F6", label: "A2" },
      { id: "D", x: 50, y: 52, color: "#EF4444", label: "D" },
      { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 38, y: 85 }, trails: [] },
    { t: 0.25, players: [
      { id: "A1", x: 38, y: 62, color: "#3B82F6", label: "A1" },
      { id: "A2", x: 75, y: 60, color: "#3B82F6", label: "A2" },
      { id: "D", x: 44, y: 52, color: "#EF4444", label: "D" },
      { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 38, y: 62 }, trails: [] },
    { t: 0.5, players: [
      { id: "A1", x: 38, y: 45, color: "#3B82F6", label: "A1" },
      { id: "A2", x: 78, y: 40, color: "#3B82F6", label: "A2" },
      { id: "D", x: 40, y: 46, color: "#EF4444", label: "D" },
      { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 38, y: 45 }, trails: [] },
    { t: 0.7, players: [

```



```

    { id: "A1", x: 38, y: 45, color: "#3B82F6", label: "A1" },
    { id: "A2", x: 78, y: 40, color: "#3B82F6", label: "A2" },
    { id: "D", x: 40, y: 46, color: "#EF4444", label: "D" },
    { id: "GK", x: 62, y: 15, color: "#EAB308", label: "GK" },
  ], ball: { x: 78, y: 40 }, trails: [{ x1: 38, y1: 45, x2: 78, y2: 40 }]
  { t: 1, players: [
    { id: "A1", x: 38, y: 45, color: "#3B82F6", label: "A1" },
    { id: "A2", x: 78, y: 40, color: "#3B82F6", label: "A2" },
    { id: "D", x: 40, y: 46, color: "#EF4444", label: "D" },
    { id: "GK", x: 62, y: 15, color: "#EAB308", label: "GK" },
  ], ball: { x: 68, y: 8 }, trails: [{ x1: 78, y1: 40, x2: 68, y2: 8, col
  ],
  staticElements: [{ type: "goal", x: 50, y: 5 }, { type: "cone", x: 38, y:
  };

```

case 7: // Crossing & Finishing

```

  return {
    fieldType: "half",
    keyframes: [
      { t: 0, players: [
        { id: "W", x: 88, y: 65, color: "#22C55E", label: "W" },
        { id: "S", x: 42, y: 42, color: "#3B82F6", label: "S" },
        { id: "CM", x: 60, y: 50, color: "#3B82F6", label: "CM" },
        { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
      ], ball: { x: 88, y: 65 }, trails: [] },
      { t: 0.3, players: [
        { id: "W", x: 88, y: 38, color: "#22C55E", label: "W" },
        { id: "S", x: 38, y: 30, color: "#3B82F6", label: "S" },
        { id: "CM", x: 55, y: 35, color: "#3B82F6", label: "CM" },
        { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
      ], ball: { x: 88, y: 38 }, trails: [] },
      { t: 0.6, players: [
        { id: "W", x: 88, y: 30, color: "#22C55E", label: "W" },
        { id: "S", x: 35, y: 22, color: "#3B82F6", label: "S" },
        { id: "CM", x: 52, y: 26, color: "#3B82F6", label: "CM" },
        { id: "GK", x: 44, y: 15, color: "#EAB308", label: "GK" },
      ], ball: { x: 88, y: 30 }, trails: [] },
      { t: 0.8, players: [
        { id: "W", x: 88, y: 30, color: "#22C55E", label: "W" },
        { id: "S", x: 35, y: 22, color: "#3B82F6", label: "S" },
        { id: "CM", x: 52, y: 26, color: "#3B82F6", label: "CM" },
        { id: "GK", x: 44, y: 15, color: "#EAB308", label: "GK" },
      ], ball: { x: 37, y: 22 }, trails: [{ x1: 88, y1: 30, x2: 37, y2: 22 }]
      { t: 1, players: [
        { id: "W", x: 88, y: 30, color: "#22C55E", label: "W" },
        { id: "S", x: 35, y: 22, color: "#3B82F6", label: "S" },
        { id: "CM", x: 52, y: 26, color: "#3B82F6", label: "CM" },

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        { id: "GK", x: 44, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 37, y: 8 }, trails: [{ x1: 37, y1: 22, x2: 37, y2: 8, col
    ],
    staticElements: [{ type: "goal", x: 50, y: 5 }, { type: "cone", x: 88, y:
    };

```

case 8: // Shadow Defending

```

    return {
        fieldType: "mini",
        keyframes: [
            { t: 0, players: [
                { id: "A", x: 30, y: 40, color: "#3B82F6", label: "A" },
                { id: "D", x: 40, y: 55, color: "#EF4444", label: "D" },
            ], ball: null, trails: [],
            label: { text: "Defender stays goal-side", x: 50, y: 88 } },
            { t: 0.2, players: [
                { id: "A", x: 55, y: 30, color: "#3B82F6", label: "A" },
                { id: "D", x: 62, y: 45, color: "#EF4444", label: "D" },
            ], ball: null, trails: [] },
            { t: 0.4, players: [
                { id: "A", x: 70, y: 55, color: "#3B82F6", label: "A" },
                { id: "D", x: 72, y: 65, color: "#EF4444", label: "D" },
            ], ball: null, trails: [] },
            { t: 0.6, players: [
                { id: "A", x: 45, y: 65, color: "#3B82F6", label: "A" },
                { id: "D", x: 50, y: 75, color: "#EF4444", label: "D" },
            ], ball: null, trails: [] },
            { t: 0.8, players: [
                { id: "A", x: 25, y: 50, color: "#3B82F6", label: "A" },
                { id: "D", x: 32, y: 62, color: "#EF4444", label: "D" },
            ], ball: null, trails: [] },
            { t: 1, players: [
                { id: "A", x: 30, y: 40, color: "#3B82F6", label: "A" },
                { id: "D", x: 38, y: 52, color: "#EF4444", label: "D" },
            ], ball: null, trails: [] },
        ],
        staticElements: [
            { type: "box", x: 10, y: 10, w: 80, h: 80 },
            { type: "label-bottom", text: "GOAL this side ↓" },
        ],
    };

```

case 9: // 3v2 Counter Attack

```

    return {
        fieldType: "half",
        keyframes: [
            { t: 0, players: [

```

```

        { id: "A1", x: 50, y: 92, color: "#3B82F6", label: "A1" },
        { id: "A2", x: 28, y: 88, color: "#3B82F6", label: "A2" },
        { id: "A3", x: 72, y: 88, color: "#3B82F6", label: "A3" },
        { id: "D1", x: 35, y: 52, color: "#EF4444", label: "D1" },
        { id: "D2", x: 65, y: 52, color: "#EF4444", label: "D2" },
        { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 50, y: 92 }, trails: [] },
    { t: 0.3, players: [
        { id: "A1", x: 50, y: 68, color: "#3B82F6", label: "A1" },
        { id: "A2", x: 18, y: 62, color: "#3B82F6", label: "A2" },
        { id: "A3", x: 78, y: 62, color: "#3B82F6", label: "A3" },
        { id: "D1", x: 35, y: 52, color: "#EF4444", label: "D1" },
        { id: "D2", x: 65, y: 52, color: "#EF4444", label: "D2" },
        { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 50, y: 68 }, trails: [] },
    { t: 0.55, players: [
        { id: "A1", x: 50, y: 52, color: "#3B82F6", label: "A1" },
        { id: "A2", x: 18, y: 45, color: "#3B82F6", label: "A2" },
        { id: "A3", x: 78, y: 45, color: "#3B82F6", label: "A3" },
        { id: "D1", x: 42, y: 48, color: "#EF4444", label: "D1" },
        { id: "D2", x: 60, y: 48, color: "#EF4444", label: "D2" },
        { id: "GK", x: 50, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 50, y: 52 }, trails: [] },
    { t: 0.72, players: [
        { id: "A1", x: 50, y: 52, color: "#3B82F6", label: "A1" },
        { id: "A2", x: 18, y: 38, color: "#3B82F6", label: "A2" },
        { id: "A3", x: 78, y: 38, color: "#3B82F6", label: "A3" },
        { id: "D1", x: 42, y: 48, color: "#EF4444", label: "D1" },
        { id: "D2", x: 60, y: 48, color: "#EF4444", label: "D2" },
        { id: "GK", x: 35, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 18, y: 38 }, trails: [{ x1: 50, y1: 52, x2: 18, y2: 38 }]
    { t: 1, players: [
        { id: "A1", x: 50, y: 52, color: "#3B82F6", label: "A1" },
        { id: "A2", x: 18, y: 30, color: "#3B82F6", label: "A2" },
        { id: "A3", x: 78, y: 38, color: "#3B82F6", label: "A3" },
        { id: "D1", x: 42, y: 48, color: "#EF4444", label: "D1" },
        { id: "D2", x: 60, y: 48, color: "#EF4444", label: "D2" },
        { id: "GK", x: 35, y: 15, color: "#EAB308", label: "GK" },
    ], ball: { x: 30, y: 8 }, trails: [{ x1: 18, y1: 30, x2: 30, y2: 8, col
    ],
    staticElements: [{ type: "goal", x: 50, y: 5 }],
};

```

```
case 10: // Switch of Play
```

```

    return {
        fieldType: "mini",
        keyframes: [

```

```

    { t: 0, players: [
      { id: "LW", x: 12, y: 35, color: "#3B82F6", label: "LW" },
      { id: "CM", x: 50, y: 50, color: "#3B82F6", label: "CM" },
      { id: "RW", x: 82, y: 35, color: "#22C55E", label: "RW" },
      { id: "D1", x: 68, y: 35, color: "#EF4444", label: "D" },
      { id: "D2", x: 72, y: 52, color: "#EF4444", label: "D" },
    ], ball: { x: 82, y: 35 }, trails: [] },
    { t: 0.25, players: [
      { id: "LW", x: 12, y: 35, color: "#3B82F6", label: "LW" },
      { id: "CM", x: 50, y: 50, color: "#3B82F6", label: "CM" },
      { id: "RW", x: 82, y: 35, color: "#22C55E", label: "RW" },
      { id: "D1", x: 68, y: 35, color: "#EF4444", label: "D" },
      { id: "D2", x: 72, y: 52, color: "#EF4444", label: "D" },
    ], ball: { x: 50, y: 50 }, trails: [{ x1: 82, y1: 35, x2: 50, y2: 50 }]
    { t: 0.5, players: [
      { id: "LW", x: 12, y: 35, color: "#3B82F6", label: "LW" },
      { id: "CM", x: 50, y: 50, color: "#3B82F6", label: "CM" },
      { id: "RW", x: 75, y: 40, color: "#22C55E", label: "RW" },
      { id: "D1", x: 62, y: 42, color: "#EF4444", label: "D" },
      { id: "D2", x: 68, y: 55, color: "#EF4444", label: "D" },
    ], ball: { x: 50, y: 50 }, trails: [] },
    { t: 0.72, players: [
      { id: "LW", x: 12, y: 35, color: "#22C55E", label: "LW" },
      { id: "CM", x: 50, y: 50, color: "#3B82F6", label: "CM" },
      { id: "RW", x: 70, y: 42, color: "#22C55E", label: "RW" },
      { id: "D1", x: 58, y: 42, color: "#EF4444", label: "D" },
      { id: "D2", x: 65, y: 55, color: "#EF4444", label: "D" },
    ], ball: { x: 12, y: 35 }, trails: [{ x1: 50, y1: 50, x2: 12, y2: 35, c
    { t: 1, players: [
      { id: "LW", x: 12, y: 22, color: "#22C55E", label: "LW" },
      { id: "CM", x: 42, y: 40, color: "#3B82F6", label: "CM" },
      { id: "RW", x: 65, y: 42, color: "#22C55E", label: "RW" },
      { id: "D1", x: 28, y: 32, color: "#EF4444", label: "D" },
      { id: "D2", x: 55, y: 50, color: "#EF4444", label: "D" },
    ], ball: { x: 12, y: 22 }, trails: [] },
  ],
  staticElements: [
    { type: "vline", x: 50 },
    { type: "label", text: "Congested", x: 72, y: 88 },
    { type: "label", text: "Space!", x: 18, y: 88 },
  ],
};

default: return null;
}
}

```

```
// — Interpolation helper —————
function lerp(a, b, t) { return a + (b - a) * t; }

function interpolateFrame(keyframes, progress) {
  if (!keyframes || keyframes.length === 0) return null;
  if (progress <= 0) return keyframes[0];
  if (progress >= 1) return keyframes[keyframes.length - 1];

  let lo = 0, hi = keyframes.length - 1;
  for (let i = 0; i < keyframes.length - 1; i++) {
    if (progress >= keyframes[i].t && progress <= keyframes[i + 1].t) {
      lo = i; hi = i + 1; break;
    }
  }
  const span = keyframes[hi].t - keyframes[lo].t;
  const localT = span === 0 ? 0 : (progress - keyframes[lo].t) / span;
  // ease in-out
  const eased = localT < 0.5 ? 2 * localT * localT : -1 + (4 - 2 * localT) * localT;

  const a = keyframes[lo], b = keyframes[hi];
  const players = a.players.map((pa) => {
    const pb2 = b.players.find(p => p.id === pa.id) || pa;
    return { ...pa, x: lerp(pa.x, pb2.x, eased), y: lerp(pa.y, pb2.y, eased) };
  });

  let ball = null;
  if (a.ball && b.ball) {
    ball = { x: lerp(a.ball.x, b.ball.x, eased), y: lerp(a.ball.y, b.ball.y, eased) };
  } else if (a.ball && eased < 0.5) ball = a.ball;
  else if (b.ball) ball = b.ball;

  const trails = eased > 0.3 ? (b.trails || []) : (a.trails || []);
  return { players, ball, trails };
}

// — Field SVG —————
function Field({ type, children, staticElements }) {
  const W = 320, H = type === "half" ? 320 : 240;
  const toX = (pct) => (pct / 100) * W;
  const toY = (pct) => (pct / 100) * H;

  return (
    <svg width={W} height={H} style={{ borderRadius: 8, display: "block", margin:
      /* Grass */
    <defs>
      <pattern id="grass" x="0" y="0" width="20" height="20" patternUnits="user
        <rect width="20" height="20" fill="#166534" />

```

```

    <rect x="0" y="0" width="10" height="20" fill="#15803d" opacity="0.4" /
</pattern>
<filter id="glow">
    <feGaussianBlur stdDeviation="2" result="coloredBlur" />
    <feMerge><feMergeNode in="coloredBlur" /><feMergeNode in="SourceGraphic"
</filter>
</defs>
<rect width={W} height={H} fill="url(#grass)" rx="8" />

{/* Static elements */}
{(staticElements || []).map((el, i) => {
    if (el.type === "cone") return (
        <polygon key={i}
            points={`${toX(el.x)},${toY(el.y) - 7} ${toX(el.x) - 5},${toY(el.y) +
            fill="#F97316" stroke="#fff" strokeWidth="0.5" />
    );
    if (el.type === "gate") return (
        <g key={i}>
            <polygon points={`${toX(el.x) - 8},${toY(el.y) - 7} ${toX(el.x) - 13}
            <polygon points={`${toX(el.x) + 8},${toY(el.y) - 7} ${toX(el.x) + 3},
            <line x1={toX(el.x) - 8} y1={toY(el.y)} x2={toX(el.x) + 8} y2={toY(el
        </g>
    );
    if (el.type === "goal") return (
        <g key={i}>
            <rect x={toX(el.x) - 38} y={toY(el.y) - 4} width={76} height={18} fil
            <rect x={toX(el.x) - 38} y={toY(el.y) - 4} width={76} height={18} fil
            <text x={toX(el.x)} y={toY(el.y) + 10} textAnchor="middle" fill="rgba
        </g>
    );
    if (el.type === "box") return (
        <rect key={i} x={toX(el.x)} y={toY(el.y)} width={toX(el.w)} height={toY
            fill="none" stroke="rgba(255,255,255,0.4)" strokeWidth="1.5" strokeDa
    );
    if (el.type === "vline") return (
        <line key={i} x1={toX(el.x)} y1={0} x2={toX(el.x)} y2={H}
            stroke="rgba(255,255,255,0.3)" strokeWidth="1.5" strokeDasharray="4,3
    );
    if (el.type === "label") return (
        <text key={i} x={toX(el.x)} y={toY(el.y)} textAnchor="middle"
            fill="rgba(255,255,255,0.5)" fontSize="9" fontStyle="italic">{el.text
    );
    if (el.type === "dashed-line") return (
        <line key={i} x1={toX(el.x1)} y1={toY(el.y1)} x2={toX(el.x2)} y2={toY(e
            stroke="rgba(255,255,255,0.2)" strokeWidth="1.5" strokeDasharray="4,3
    );
    return null;

```

```

    }}}

    { /* Field lines for half-field */
    { type === "half" && (
      <>
        <rect x={2} y={2} width={W-4} height={H-4} fill="none" stroke="rgba(255
        <rect x={toX(25)} y={toY(0)} width={toX(50)} height={toY(22)} fill="non
        </>
      )}
    { type === "mini" && (
      <rect x={4} y={4} width={W-8} height={H-8} fill="none" stroke="rgba(255,2
    )}

    {children}
  </svg>
);
}

```

```

// — Animated Drill Canvas —————
function DrillCanvas({ drill, isPlaying, progress }) {
  const anim = getDrillAnimation(drill.id);
  if (!anim) return null;

  const W = 320, H = anim.fieldType === "half" ? 320 : 240;
  const toX = (pct) => (pct / 100) * W;
  const toY = (pct) => (pct / 100) * H;

  const frame = interpolateFrame(anim.keyframes, progress);
  if (!frame) return null;

  return (
    <Field type={anim.fieldType} staticElements={anim.staticElements}>
      { /* Pass trails */
        (frame.trails || []).map((tr, i) => {
          const dx = toX(tr.x2) - toX(tr.x1), dy = toY(tr.y2) - toY(tr.y1);
          const len = Math.sqrt(dx*dx+dy*dy);
          const ux = dx/len, uy = dy/len;
          const arrowX = toX(tr.x2) - ux*8, arrowY = toY(tr.y2) - uy*8;
          const angle = Math.atan2(dy, dx);
          const ax1 = arrowX - 8*Math.cos(angle-0.5), ay1 = arrowY - 8*Math.sin(ang
          const ax2 = arrowX - 8*Math.cos(angle+0.5), ay2 = arrowY - 8*Math.sin(ang
          return (
            <g key={i}>
              <line x1={toX(tr.x1)} y1={toY(tr.y1)} x2={toX(tr.x2)} y2={toY(tr.y2)}
                stroke={tr.color || "#FCD34D"} strokeWidth="2.5" strokeLinecap="rou
              <polygon points={` ${toX(tr.x2)}, ${toY(tr.y2)} ${ax1}, ${ay1} ${ax2}, ${
                fill={tr.color || "#FCD34D"} opacity="0.85" />
            </g>
          )
        })
      }
    </Field>
  );
}

```

```

        </g>
    );
}}}

{/* Players */}
{(frame.players || []).map((p) => (
    <g key={p.id}>
        <circle cx={toX(p.x)} cy={toY(p.y)} r={13} fill={p.color} stroke="white"
            style={{ filter: "drop-shadow(0 2px 4px rgba(0,0,0,0.5))" }} />
        <text x={toX(p.x)} y={toY(p.y) + 4} textAnchor="middle"
            fill="white" fontSize="9" fontWeight="bold" fontFamily="monospace">{p
        </g>
    )})

{/* Ball */}
{frame.ball && (
    <g>
        <circle cx={toX(frame.ball.x)} cy={toY(frame.ball.y)} r={6}
            fill="white" stroke="#333" strokeWidth="1.5"
            style={{ filter: "drop-shadow(0 1px 3px rgba(0,0,0,0.6))" }} />
        {/* Ball pattern */}
        <circle cx={toX(frame.ball.x)} cy={toY(frame.ball.y)} r={3}
            fill="#333" opacity="0.3" />
    </g>
)}
</Field>
);
}

```

```

// — Legend —————
function Legend() {
    return (
        <div style={{ display: "flex", gap: 16, flexWrap: "wrap", justifyContent: "ce
        {[
            { color: "#3B82F6", label: "Player" },
            { color: "#EF4444", label: "Defender" },
            { color: "#22C55E", label: "Wide / Free" },
            { color: "#EAB308", label: "Goalkeeper" },
        ]}.map(({ color, label }) => (
            <div key={label} style={{ display: "flex", alignItems: "center", gap: 6 }
                <div style={{ width: 12, height: 12, borderRadius: "50%", background: c
                <span style={{ fontSize: 11, color: "rgba(255,255,255,0.6)", fontFamily
                </div>
            )})
        <div style={{ display: "flex", alignItems: "center", gap: 6 }}>
            <div style={{ width: 20, height: 2, background: "#FCD34D", borderRadius:
            <span style={{ fontSize: 11, color: "rgba(255,255,255,0.6)", fontFamily:

```



```

        </div>
    </div>
  );
}

// — Main App —————
export default function SoccerDrills() {
  const [activeDrill, setActiveDrill] = useState(0);
  const [isPlaying, setIsPlaying] = useState(false);
  const [progress, setProgress] = useState(0);
  const [stepIndex, setStepIndex] = useState(0);
  const rafRef = useRef(null);
  const startRef = useRef(null);
  const pausedAtRef = useRef(0);
  const DURATION = 4000; // ms per loop

  const drill = DRILLS[activeDrill];

  const animate = useCallback((ts) => {
    if (!startRef.current) startRef.current = ts - pausedAtRef.current * DURATION
    const elapsed = ts - startRef.current;
    const p = (elapsed % DURATION) / DURATION;
    setProgress(p);
    const steps = drill.steps || [];
    setStepIndex(Math.min(Math.floor(p * steps.length), steps.length - 1));
    rafRef.current = requestAnimationFrame(animate);
  }, [drill]);

  useEffect(() => {
    if (isPlaying) {
      rafRef.current = requestAnimationFrame(animate);
    } else {
      if (rafRef.current) cancelAnimationFrame(rafRef.current);
      pausedAtRef.current = progress;
    }
    return () => { if (rafRef.current) cancelAnimationFrame(rafRef.current); };
  }, [isPlaying, animate]);

  // Reset on drill change
  useEffect(() => {
    setIsPlaying(false);
    setProgress(0);
    setStepIndex(0);
    startRef.current = null;
    pausedAtRef.current = 0;
  }, [activeDrill]);

```

```

const pillars = ["All", "Dribbling", "Passing", "Finishing", "Defending"];
const [filterPillar, setFilterPillar] = useState("All");
const filtered = DRILLS.filter(d => filterPillar === "All" || d.pillar === filt

return (
  <div style={{
    minHeight: "100vh",
    background: "linear-gradient(135deg, #0a1a0f 0%, #0d2416 40%, #0a1a0f 100%)",
    fontFamily: "'Courier New', monospace",
    color: "white",
    padding: "0 0 40px",
  }}>
    { /* Header */ }
    <div style={{
      background: "linear-gradient(90deg, #14532d, #166534, #14532d)",
      borderBottom: "2px solid #22c55e",
      padding: "20px 24px",
      display: "flex",
      alignItems: "center",
      gap: 16,
    }}>
      <div style={{ fontSize: 36 }}>⚽</div>
      <div>
        <div style={{ fontSize: 22, fontWeight: "bold", letterSpacing: 2, color
          U10 DRILL ANIMATOR
        </div>
        <div style={{ fontSize: 11, color: "rgba(134,239,172,0.6)", letterSpacing: 2,
          8v8 SEASON COACHING PLAN — 10 INTERACTIVE DRILLS
        </div>
      </div>
    </div>

    <div style={{ display: "flex", gap: 0, height: "calc(100vh - 90px)", overfl

      { /* Sidebar */ }
      <div style={{
        width: 220,
        background: "rgba(0,0,0,0.4)",
        borderRight: "1px solid rgba(34,197,94,0.2)",
        overflowY: "auto",
        flexShrink: 0,
      }}>
        { /* Filter */ }
        <div style={{ padding: "12px 10px 8px", borderBottom: "1px solid rgba(2
          <div style={{ fontSize: 9, color: "rgba(255,255,255,0.4)", marginBott
            <div style={{ display: "flex", flexWrap: "wrap", gap: 4 }}>
              {pillars.map(p => (

```

```

        <button key={p} onClick={() => setFilterPillar(p)} style={{
          padding: "3px 8px", fontSize: 9, borderRadius: 99, cursor: "pointer",
          background: filterPillar === p ? "#22c55e" : "rgba(255,255,255,0.6)",
          color: filterPillar === p ? "#000" : "rgba(255,255,255,0.6)",
          fontWeight: filterPillar === p ? "bold" : "normal",
          letterSpacing: 0.5,
        }}>{p}</button>
      </div>
    </div>

    {/* Drill list */}
    {filtered.map((d) => {
      const realIdx = DRILLS.findIndex(x => x.id === d.id);
      const active = realIdx === activeDrill;
      return (
        <div key={d.id} onClick={() => setActiveDrill(realIdx)} style={{
          padding: "10px 12px",
          cursor: "pointer",
          background: active ? "rgba(34,197,94,0.15)" : "transparent",
          borderLeft: active ? "3px solid #22c55e" : "3px solid transparent",
          borderBottom: "1px solid rgba(255,255,255,0.05)",
          transition: "all 0.15s",
        }}>
          <div style={{ display: "flex", alignItems: "center", gap: 8, margin: 0 }}>
            <div style={{
              width: 20, height: 20, borderRadius: "50%",
              background: d.pillarColor, display: "flex", alignItems: "center",
              fontSize: 9, fontWeight: "bold", flexShrink: 0,
            }}>{d.id}</div>
            <div style={{ fontSize: 11, fontWeight: active ? "bold" : "normal",
              {d.name}
            }}>
              <div>
                <div style={{ fontSize: 9, color: d.pillarColor, marginLeft: 28,
                  {d.pillar} · {d.duration}
                }}>
              </div>
            </div>
          </div>
        );
      )}
    </div>

    {/* Main panel */}
    <div style={{ flex: 1, overflowY: "auto", padding: "20px 24px" }}>

      {/* Drill header */}
      <div style={{ display: "flex", alignItems: "flex-start", justifyContent: "space-between", padding: "10px 0" }}>

```

```

<div>
  <div style={{ display: "flex", alignItems: "center", gap: 10, margin: 10 }}>
    <div style={{ width: 28, height: 28, borderRadius: "50%", background-color: {drill.id} }}>
    </div>
    <h2 style={{ margin: 0, fontSize: 20, color: "#86efac", letterSpacing: 1 }}>
    </div>
    <div style={{ display: "flex", gap: 12, flexWrap: "wrap" }}>
      {[`${🕒` ${drill.duration}`, `👥` ${drill.players}`, `🎯` ${drill.focus}]}.map((item, index) => {
        <span key={tag} style={{ fontSize: 11, color: "rgb(255,255,255)" }}>{item}</span>
      })
    </div>
  </div>
  <div style={{
    background: "rgb(34,197,94,0.1)", border: "1px solid rgb(34,197,94,0.2)",
    borderRadius: 8, padding: "8px 14px", maxWidth: 260,
  }}>
    <div style={{ fontSize: 9, color: "#22c55e", letterSpacing: 1, margin: 0 }}>
      <div style={{ fontSize: 12, color: "#86efac", fontStyle: "italic" }}>
      </div>
    </div>
  </div>

  {/ * Animation area */}
  <div style={{ display: "flex", gap: 20, flexWrap: "wrap" }}>
    <div style={{ flex: "0 0 auto" }}>
      <div style={{
        background: "rgb(0,0,0,0.3)",
        border: "1px solid rgb(34,197,94,0.2)",
        borderRadius: 12,
        padding: 16,
      }}>
        <DrillCanvas drill={drill} isPlaying={isPlaying} progress={progress} />
        <Legend />
      </div>
    </div>

    {/ * Controls */}
    <div style={{ display: "flex", alignItems: "center", gap: 12, margin: 10 }}>
      <button onClick={() => { setProgress(0); startRef.current = null }} style={{
        background: "rgb(255,255,255,0.08)", border: "1px solid rgb(34,197,94,0.2)",
        color: "white", padding: "7px 14px", borderRadius: 8, cursor: "pointer"
      }}>🔄 Reset</button>

      <button onClick={() => setIsPlaying(p => !p)} style={{
        background: isPlaying ? "rgb(239,68,68,0.2)" : "rgb(34,197,94,0.1)",
        border: `1px solid ${isPlaying ? "#ef4444" : "#22c55e"}`,
        color: "white", padding: "7px 22px", borderRadius: 8, cursor: "pointer",
        fontWeight: "bold", letterSpacing: 1,
      }}>
    </div>
  </div>

```

```

        {isPlaying ? "⏸ Pause" : "▶ Play"}
      </button>
    </div>

    {/* Progress bar */}
    <div style={{ marginTop: 12, height: 4, background: "rgba(255,255"
      <div style={{
        height: "100%", width: `${progress * 100}%`,
        background: "linear-gradient(90deg, #22c55e, #86efac)",
        borderRadius: 2, transition: "width 0.05s linear",
      }} />
    </div>
  </div>
</div>

{/* Steps + Info panel */}
<div style={{ flex: 1, minWidth: 200 }}>
  {/* Step by step */}
  <div style={{
    background: "rgba(0,0,0,0.25)", border: "1px solid rgba(34,197,94"
    borderRadius: 12, padding: 16, marginBottom: 16,
  }}>
    <div style={{ fontSize: 10, color: "#22c55e", letterSpacing: 1, m
    {drill.steps.map((step, i) => {
      const isActive = i === stepIndex && isPlaying;
      const isPast = i < stepIndex && isPlaying;
      return (
        <div key={i} style={{
          display: "flex", alignItems: "flex-start", gap: 10, marginB
          opacity: isPlaying ? (isPast ? 0.4 : isActive ? 1 : 0.5) :
          transition: "opacity 0.3s",
        }}>
          <div style={{
            width: 22, height: 22, borderRadius: "50%", flexShrink: 0
            background: isActive ? "#22c55e" : isPast ? "#166534" : "
            border: `1.5px solid ${isActive ? "#86efac" : "rgba(255,2
            display: "flex", alignItems: "center", justifyContent: "c
            fontSize: 10, fontWeight: "bold",
            color: isActive ? "#000" : "rgba(255,255,255,0.6)",
            transition: "all 0.3s",
          }}>{i + 1}</div>
          <div style={{
            fontSize: 12, color: isActive ? "#86efac" : "rgba(255,255
            paddingTop: 3, lineHeight: 1.4, fontWeight: isActive ? "b
            transition: "color 0.3s",
          }}>{step}</div>
        </div>
      )
    })}
  </div>

```

```

        );
    }}}
</div>

{/* Navigation */}
<div style={{
    background: "rgba(0,0,0,0.25)", border: "1px solid rgba(255,255,2
    borderRadius: 12, padding: 14,
}}>
    <div style={{ fontSize: 10, color: "rgba(255,255,255,0.4)", lette
    <div style={{ display: "flex", gap: 8, flexWrap: "wrap" }}>
        {DRILLS.map((d, i) => (
            <button key={d.id} onClick={() => setActiveDrill(i)} style={{
                width: 32, height: 32, borderRadius: "50%", cursor: "pointe
                background: i === activeDrill ? d.pillarColor : "rgba(255,2
                color: i === activeDrill ? "#000" : "rgba(255,255,255,0.6)"
                fontSize: 11, fontWeight: "bold",
            }}>{d.id}</button>
        ))}
    </div>
    <div style={{ display: "flex", gap: 8, marginTop: 12 }}>
        <button onClick={() => setActiveDrill(i => Math.max(0, i - 1))}
            flex: 1, padding: "8px 0", borderRadius: 8, cursor: "pointer"
            background: activeDrill === 0 ? "rgba(255,255,255,0.04)" : "r
            color: activeDrill === 0 ? "rgba(255,255,255,0.2)" : "white",
        }}>← Prev</button>
        <button onClick={() => setActiveDrill(i => Math.min(DRILLS.leng
            flex: 1, padding: "8px 0", borderRadius: 8, cursor: "pointer"
            background: activeDrill === DRILLS.length - 1 ? "rgba(255,255
            color: activeDrill === DRILLS.length - 1 ? "rgba(255,255,255,
            fontSize: 13, fontWeight: "bold",
        }}>Next →</button>
    </div>
</div>
</div>
</div>
</div>
</div>
</div>
);
}

```